

SHLOK JAIN

Second-Year Undergraduate

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[GitHub](#) | [LinkedIn](#) | [Codeforces](#) | [AtCoder](#)

Academic Qualifications

Year	Degree/Certificate	Institute/Administrator	GPA/%
2024 - Present	B.E. CS + M.Sc. Economics	Birla Institute of Technology & Science, Pilani	8.81/10
2024	State Board (XII)	Dixit Road High School, Mumbai	89%

Key Projects & Experiences

- Kaze** (May'21 – Jun'21)
 - Developed a productivity-focused Android launcher using **Flutter** and **SQLite** for on-device persistence and **Firebase** for analytics.
 - Designed the end-to-end product including user flows, screens, wireframes, and design systems for multiple unique devices on **Figma**.
 - Achieved **2K+ beta downloads** through organic user adoption, maintaining a strong **4.2/5.0** average rating.
 - Placed **2nd** in India's Future Tycoons and received a **seed fund of 50K rupees**.
- Open-Source Contributor @ HPX (Parallel C++ Runtime)** (Jul'25 – Present)
 - Actively contributing to HPX, **owning the Sender/Receiver infrastructure** which abstracts async task execution cleanly.
 - Merged [PR #6734](#), resolving incorrect and redundant `resolver_client` aliases in `agas/agas_fwd.hpp`.
 - Merged [PR #6899](#) in HPX **implementing forwarding_sender_query** across core sender adaptors, establishing correct query propagation through composed sender chains.
- Pollu** (Dec'25 – Jan'26)
 - Built a **30 m spatial resolution** public API to **predict hourly AQI** by designing an ML model using ground sensors and satellite data, achieving **error of 14 units** and an **R² (coefficient of determination) of 0.7**, solo in one month.
 - Engineered scalable ETL pipelines **processing 30 GB of raw data**, automating ingestion, cleaning, and feature assembly for downstream machine learning workflows.
 - Implemented a **high-throughput LightGBM inference engine** and durable persistence via queued writes, connection pooling and fault tolerant workers with automatic restart and idempotent guarantees to ensure reliable end to end processing.
 - Architected a **distributed Go backend** implementing **request deduplication** through leader-follower, multi-tier **caching** (Redis hot cache + DuckDB cold cache) which had a **hit rate of 99.8%**. Improved p95 latency from 900ms to 200ms.
 - Designed and implemented a minimal, responsive user interface using **JavaScript and CSS**, allowing users to drop pins anywhere on the map to retrieve real-time AQI values, with clear hazard categorization and intuitive visual cues.
- Engineering & Operations @ Soma Insurance (Backed by Accel)** (March '26 - Present)
 - Owned the **infrastructure** running our entire fleet of AI agents. A distributed session runtime on a **zero-downtime FastAPI** server sustaining **500k req/day** peak with **concurrent session pools**, global **backpressure**, **idempotent retries**, orphan session handling, **crash recovery** and tiered failure handling, and **e2e observability routing** for unexpected crashes.
 - Owned **policy quote turnaround time** as the metric by building **browser automation** that submits applications across portals in the background, cutting turnaround **from 1 day to 30 min (48x)** at a **concurrency of 100+ clients** that previously bottlenecked on manual submission.
 - Designed the interface** used to supervise agentic operations, custom-built for each specific workflow to surface **live session state**, **escalate exceptions**, and **abort runs in-flight**. Helped transition a fully manual process into a semi-autonomous one requiring only human approvals.
 - Joined live sales calls and **learned everything about insurance operations** from licensed brokers to map a particular **manual workflow: submissions**. This involved pristine **industry form-filling** with the appropriate business details, **optimally matching** which carriers to hit for which types of business according to their risk. **Fully automated** this workflow in a team of 2 by deploying specific skills and MCPs for each task.

Technical Skills & Relevant Courses

Programming Languages	C++ — Go — Python — JS — TypeScript — Dart — SQL
Frameworks/Libraries	React — FastAPI — Flutter — HPX — node.js — ExpressJS
Build Systems/Tooling	CMake — Git — npm
Relevant Courses	Multivariable Calculus (A) — Linear Algebra & Complex Systems (A) — Computer Programming (A) — Fundamentals of Finance & Accounting (A) — Derivatives & Risk Management (*)

Positions of Responsibility

- Competitive Programmer, Coding Club** (Feb'25 - Present)
 - Rated **1554** (peak) on [Codeforces](#), with the best **rank of 994** out of 30K participants in Round 1033 (Div. 2).
 - Rated **1104** (peak) on [AtCoder](#), with the best **rank of 695** out of 8k participants in the AtCoder Beginner Contest 423.
- Teaching Assistant, Computer Programming** (Jan'26 - Present)
 - Conducting of laboratory sessions, mentoring students, and clarifying core concepts during hands-on coursework.
 - Responsible for setting and evaluating examinations and assessments for a cohort of **600+ students**.

Extra-Curricular Activities

- District-level Handball player in Mumbai circuits
- Participated in district-level elocution competitions
- Qualified IOQM, first stage of Indian mathematics olympiad (equivalent to AMC 12)